Metagenomics workshop

Pre-workshop: Try to carry out Step 1 due to time constraints

Metagenomics is the study of all organisms in an environment. In previous years, 16S rRNA genes were used to identify bacteria from environments using PCR amplicons which were then sequenced. A large set of targeted software has grown up to deal with 16S results. Whilst 16S is still somewhat used due to its cost effectiveness, whole metagenome sequencing is a growing field that is viewed as less biased.

For this workshop you have access to a microbial community dataset which consists of a mixed community of different microbial strains in different abundances. It is a small community dataset that was sequenced by Illumina short read DNA sequencing. This dataset can be used throughout sequencing labs to calibrate sequencing procedures, build and assess the performance of metagenomic workflows.

The aim of our study (questions to answer):
What genus or species of microbial strains are present?
What abundances are they found in?

Tutorial:
Due to high compute processing and time requirement, a metagenomic assembly has been performed for you using the metagenomic assembler Megahit[1].

Input Files:
Your first input file is the metagenomic assembly CommunityScaffolds.fasta. This consists of a large set of contigs from all the organisms in the metagenome. Your task is to separate them by binning using two different binning programs.
Binning is the process of separating different microbial genomes from all the other and putting them into different bins, one per genome. We will then check the quality of the bins and finally classify them into genus and species.

Step 1:
Start an interactive job:

```
Interactive-bio-ds
```

```
cd scratch/
```

Git clone the metagenomics workshop tutorial from:

```
git clone https://github.com/LCrossman/Metagenomics_shotgun_tutorial.git
```

```
cd Metagenomics_shotgun_tutorial
cd ReadAligns
```

Next:

```
sbatch readalign_bam_and_sort.sh
```

This script will copy the illumina paired-end reads EDME*.fq.gz (forward R1 and reverse R2) to your scratch and align them to the metagenome assembly CommunityScaffolds.fasta (which you can find in the RawData folder of the Metagenomics_shotgun_tutorial. If you prefer you can align the reads according to your previous lab on read alignments, using either the program bwa[2] or bowtie2[3] (**not** STAR, it's for RNA).
The reads are located in /gpfs/BIO-DSB/Session7/MG_workshop_2026/FASTQ/
and will be first copied to your scratch directory inside the sbatch script.
If you have difficulty with the sbatch script relating to copying the read files or you want to have them on a local computer you can also access them at:
https://shorturl.at/dkXaj

alternatively you can copy them to your local computer using (on one line):

```
scp /gpfs/BIO-DSB/Session7/MG_workshop_2026/FASTQ/EDME*fq.gz
<username>@hali.uea.ac.uk:/where/you/want/it
```

Reads are big data and approximately 2Gb each.

The final output of this script needs to be a sorted and index bam file (Community_aligned.bam.sorted.bam)

The sorted bam file contains information about how many reads have aligned to which contigs.
Note this will take a few minutes to run because it is quite a compute intensive process.
If using bowtie2[3], (like in the `sbatch` script `readalign_bam_and_sort.sh` above), after the read alignment stage, the alignment statistics are printed. These may be found in your Error or Output messages. You may like to copy these and save to a text file for later consideration.

output files:

```
Community_aligned.bam.sorted.bam
Community_aligned.bam.sorted.bam.bai
```

You will need these files to move on to the next step.

Next we move onto
Step 2 Binning:
You will need to be inside the Binning folder of Metagenomics_shotgun_tutorial

```
cd ..
cd Binning
```

In this folder there are two binning scripts – one is for an older but very tried and tested traditional method of binning using metabat2[4], which uses sequence characteristics like tetranucleotide patterns and coverage depth to carry out binning.
The other is for a new binning method, SemiBin2[5] which uses AI to separate the contigs into bins.
Step 2.1 metabat2 binning:

```
sbatch binning_workflow.sh
```

Note: if you receive a message that

```
Found conda-prefix at '/gpfs/home/USERNAME/.conda/envs/unpackenv'.
Overwrite? [y/N]
```

You can safely select `y`

In the sbatch script, we will first create a new input file `depth.txt` from the bam files which is result of calculating the coverage depth over the metagenomic assembly.
The next specific binning command is best for simple communities. This uses a default minContig size 2000, a parameter that may need to be reduced for more complex communities.

Metabat2 output:
You can check on the output progress by reading the file at

```
~/scratch/Data/Metagenomics_shotgun_tutorial/Binning/Output_messages/Metagenomics_tutorial*.out
```

When complete, you should see files:

```
depth.txt
binning.tsv
```

among some others
What we need is the bin files (`*fasta`) located at your output directory, and the refined bins which are found in refined_bins in your output directory.
These will be picked up by the sbatch script in step 3.

Step 2.2: SemiBin2

```
sbatch ML_binning_workflow.sh
```

After the run is complete, you should see a directory semibinout.
There are several intermediate files which may be of interest to your group project.
The final bin files are found in the output_bins directory inside semibinout

Step 3 Quality assessment of the bins – using CheckM[6]
We need to check the quality of the bins in terms of contamination (does the bin genome file contain more than one organism?) and completeness (does the bin genome file contain all the expected housekeeping genes?).

```
cd ..
cd CheckQual
sbatch checkm_quality.sh
```

Your output file will be `checkm_result.tsv`

Identify your best bins in terms of completeness (you want high completeness as %) and contamination (you want low contamination as %).

CheckM provides some data about taxonomy but you will get better resolution from the dedicated classification program in the next step. The statistics for completeness, contamination and Strain heterogeneity are what we want here.

Step 4. Classification

Now take your top 10 bins (from the quality check above) and identify the genus/species with:

```
cd ..
cd Classify
sbatch taxa_classify_bins.sh
```

What species did you identify? Did you identify any issues? Why or Why not?

Further/Group work:

Whilst MetaBat2 is highly performant, there are newer binning programs available, mainly using machine learning/deep learning AI, including SemiBin2.

Either:

You can compare the results from the two metagenomic binning programs (and also the refined bins) to see which performs the best and investigate some alternative parameters. You can find more about alternative parameters for metabat2 in the software manual documentation. You can also investigate the R script in Optional plotting of Metagenomics_shotgun_tutorial. The files depth.txt and binning.txt are generated by metabat2 during the run.

Or:

Here is a link to access some long-read Oxford Nanopore data to integrate and see how that improves binning performance. You can use the reads directly in the binning process as if they are contigs from a metagenome assembly. You can join them together with the assembly contigs using cat.

https://shorturl.at/BjCHo

Or:

Here is a link to some 'real world' data from a professor's garden: There are three samples – can you find any differences in the bins across the samples? This is a research project.

https://shorturl.at/VZtAJ

What parameters did you consider in choosing the 'best' binning strategy and result?

1. Li, D., Liu, C.-M., Luo, R., Sadakane, K. & Lam, T.-W. MEGAHIT: an ultra-fast single-node solution for large and complex metagenomics assembly via succinct de Bruijn graph. *Bioinformatics* **31**, 1674–1676 (2015).
2. Li, H. Aligning sequence reads, clone sequences and assembly contigs with BWA-MEM. *arXiv preprint arXiv* **00**, 3 (2013).
3. Langmead, B. & Salzberg, S. L. Fast gapped-read alignment with Bowtie 2. *Nat Methods* **9**, 357–359 (2012).
4. Kang, D. D. *et al.* MetaBAT 2: an adaptive binning algorithm for robust and efficient genome reconstruction from metagenome assemblies. *PeerJ* **7**, e7359 (2019).
5. Pan, S., Zhao, X.-M. & Coelho, L. P. SemiBin2: self-supervised contrastive learning leads to better MAGs for short- and long-read sequencing. *Bioinformatics* **39**, i21–i29 (2023).

6. Parks, D. H., Imelfort, M., Skennerton, C. T., Hugenholtz, P. & Tyson, G. W. CheckM: Assessing the quality of microbial genomes recovered from isolates, single cells, and metagenomes. *Genome Res.* **25**, (2015).